

A ground source refrigeration and air conditioning system with cross-seasonal energy storage capability

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Abstract

To make use of shallow geothermal energy, Ground Source Refrigeration and Air Conditioning (GSRAC) systems becomes more and more popular. However, the performance of a conventional GSRAC with a borehole heat exchanger (BHE) may deteriorate over years due to the soil thermal imbalance issue that affects these systems. This study proposes a novel concept of using soil as media for seasonal cold energy storage via a passive device termed as Thermal Diode Tank (TDT). The TDT consists of an insulated water tank fitted with an array of heat pipes. By integrating the TDT into a conventional GSRAC system, “cold” energy can be passively collected from ambient air during winter, injected into the BHE, and stored in the soil. The stored “cold” energy can then be retrieved in the summer, facilitating the cross-seasonal cold energy storage (CS). Thus, a conventional GSRAC system can be transformed into a GSRAC system with cross-seasonal cold energy storage capability, ie. GSRAC+CS system. The validated BHE model previously developed by the authors is used to predict the performance improvements achieved by the GSRAC+CS system. Results indicate that the Annual Net Cold Energy Storage Efficiency (ANESE), increased from 5.7% to 10.7% over a ten-year period. The average Borehole Performance Improvement (BPI) due to the addition of cold storage capability is 11% over the same timeframe. The study also discusses the impacts of varying design and operational parameters on ANESE and BPI. Results demonstrated that GSRAC+CS systems not only mitigate the soil thermal imbalance issue faced by conventional GSRAC systems but also require less BHE depth to achieve equivalent performance.